

UseEffect

1.What is useEffect?

useEffect is used to perform tasks that are **outside the normal rendering process**, like:

- Fetching data from an API
- Updating the DOM manually
- Setting up timers
- Subscribing to events

"Run some code after the component renders"

Basic Syntax

```
import { useEffect } from "react";
```

```
useEffect(() => {  
  // side-effect code  
  
  return () => {  
    // cleanup (optional)  
  };  
}, [dependencies]);
```

Where do we use useEffect?

1. API Call (Most Common)

```
useEffect(() => {  
  fetch("https://api.example.com/data")  
    .then(res => res.json())  
    .then(data => console.log(data));  
}, []);
```

Runs **only once** (on mount)

2.Timer / Interval

```
useEffect(() => {  
  const timer = setInterval(() => {  
    console.log("Running...");  
  }, 1000);  
}, [dependencies]);
```

```
return () => clearInterval(timer); // cleanup
}, []);
```

3.Event Listener

```
useEffect(() => {
  const handleResize = () => console.log(window.innerWidth);

  window.addEventListener("resize", handleResize);

  return () => window.removeEventListener("resize", handleResize);
}, []);
```

4.When State Changes

```
useEffect(() => {
  console.log("Count changed");
}, [count]);
```

Runs **every time count changes**

Dependency Array

Dependency	Behavior
[]	Run only once (component mount)
[count]	Run when count changes
No array	Run on every render (avoid usually)

Real Life Example

```
import { useState, useEffect } from "react";

function App() {
  const [count, setCount] = useState(0);

  useEffect(() => {
    console.log("Component mounted or updated");
  }, [count]);

  return (
```

```
<button onClick={() => setCount(count + 1)}>
  {count}
</button>
);
}
```

- useEffect = **side effects handle karne ke liye**
- Runs **after render**
- Controlled by **dependency array**